

UMC (2303 TT)

PIC/TFLN Upside Unlocks a Higher-Margin Cycle

Maintained **Buy**
Target price NT\$160

Lift TP to NT\$160: UMC's share price underperformed the Taiex by 26% in 2H-July, due to TSMC's comment of "mature nodes not in shortage" but has moved sideways month-to-date to digest the company's higher depreciation guidance and earlier crowded long position. Nevertheless, we now expect the acceleration of its high-margin PIC business, mainly driven by Hyperlight (which could have made significant progress in Google) and TeraHop. We expect the wafer shipments to grow to 20k/month and 40k/month by 4Q27 and 4Q28, with gross margin being significantly higher than corporate levels. Additionally, we expect its UTR to continue to move higher, driven by AI (incl. optics)/CIS/ISP and DDIC which benefits from Apple's move its Mac from 8" to 55nm. While the market has less appetite on price-hike theme, we believe the visibility has been extended to 2H27. Regarding Intel, mgmt earlier reiterated that the collaboration remains on track for tape-out in 2027 on Intel 12, and we expect the capacity to be more than one phrase. Incorporating higher pricing and PIC demand, we raise 2026E/2027E EPS estimates by 2%/14% and move target multiple higher to 20x on higher margin outlook, resulting in a new TP of NT\$160.

TFLN PIC demand to accelerate: Driven by the technology migration to 400G per lane, we believe that UMC's TFLN PIC business, mainly 8", has seen a sharp upward revision. We estimate the wafer shipments to grow to 20k/month and 40k/month by 4Q27 and 4Q28, from 4-5k/month by 4Q26. Based on >70% gross margin and \$2.5k/8" ASP (vs. normal 8" ASP of below \$1k), we expect EPS contribution to be \$0.7/\$1.5 in 2027/2028. We attribute the robust order visibility to HyperLight, which could have made significant progress with Google. We also expect solid demand from TeraHop, which is driven by Nvidia and Google. Due to industry-wide supply tightness and manufacturing complexity, we believe TFLN is a high-margin business, helping UMC shift its product mix toward higher-value AI (incl. optics) segment. Technology wise, we don't rule out the possibility for UMC to extend its PIC wafer business to packaging level in the mid/long term.

Risks: 1) Geo-political risk; 2) Demand uncertainties; 3) Competition; 4) New product delay.

Profit forecast

	2024	2025	2026E	2027E	2028E
Revenue (m)	232,303	237,553	290,217	353,625	406,835
Revenue YoY (%)	4.4	2.3	22.2	21.8	15.0
Net profit (m)	47,211	41,716	92,701	99,138	120,772
Net profit YoY (%)	-22.6	-11.6	122.2	6.9	21.8
EPS (NT\$)	3.8	3.3	7.4	7.9	9.7
P/E	19.6	22.3	10.0	9.4	7.7
ROE (%)	12.5	11.0	21.2	21.5	21.2

Source: Company data, GF Securities (Hong Kong) Brokerage.

Please be sure to read the disclosures at the end of this report carefully.

Jeff Pu, CFA

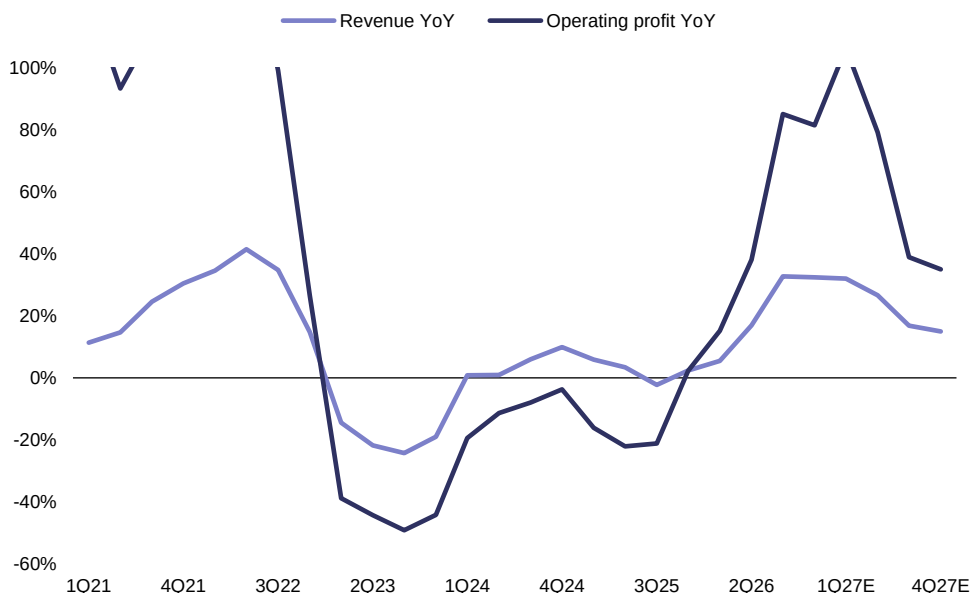
SFC CE No. BNO719
jeffpu@gfgroup.com.hk

Henry Huang

SFC CE No. BYA749
henryhuang@gfgroup.com.hk

Earnings Forecast

Figure 1: UMC's revenue YoY and operating profit YoY (%)



Sources: Company data, GF Securities (Hong Kong) Brokerage

Earnings Revision

Incorporating higher pricing and PIC demand, we raise 2026E/2027E EPS estimates by 2%/14%.

Figure 2: Earnings revision

NTD m	2025	2026E		2027E		Change (%)	
		Old	New	Old	New	2026E	2027E
Net sales	237,553	286,813	290,217	338,898	353,625	1%	4%
Gross profit	68,906	96,217	98,533	126,859	141,264	2%	11%
EBIT	43,949	66,798	69,014	94,861	108,706	3%	15%
Pre-tax Income	49,648	99,454	101,828	101,935	116,265	2%	14%
Net profit	41,716	90,672	92,701	86,824	99,138	2%	14%
EPS (NT\$)	3.34	7.26	7.43	6.95	7.94	2%	14%
Key ratios (%)							
Gross margin	29.0%	33.5%	34.0%	37.4%	39.9%		
Operating margin	18.5%	23.3%	23.8%	28.0%	30.7%		
Pre-tax margin	20.9%	34.7%	35.1%	30.1%	32.9%		
Net profit margin	17.6%	31.6%	31.9%	25.6%	28.0%		

Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 3: UMC's P&L forecast

(NTD m)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
Sales	237,553	61,038	68,733	78,537	81,910	290,217	80,585	87,042	91,783	94,214	353,625
COGS	168,647	43,219	46,410	50,210	51,845	191,684	49,633	52,226	54,824	55,678	212,361
Gross profit	68,907	17,818	22,323	28,327	30,065	98,533	30,952	34,817	36,959	38,537	141,264
Opex	24,959	6,542	7,373	7,737	7,867	29,519	7,667	8,007	8,337	8,547	32,558
Operating profit	43,948	11,276	14,950	20,590	22,198	69,014	23,285	26,810	28,622	29,990	108,706
Non-op profit	5,699	5,367	30,236	-4,545	1,755	32,814	1,890	1,890	1,890	1,890	7,559
Pre-tax profit	49,647	16,643	45,186	16,045	23,953	101,828	25,175	28,699	30,512	31,879	116,265
Income tax	8,113	527	2,962	2,327	3,473	9,289	3,776	4,448	4,424	4,622	17,271
Minority interest	182	54	36	36	36	162	36	36	36	36	144
Net Income	41,717	16,171	42,260	13,755	20,516	92,701	21,435	24,287	26,123	27,293	99,138
EPS (NT\$)	3.34	1.29	3.39	1.10	1.64	7.43	1.72	1.95	2.09	2.19	7.95
Outstanding shares	12,485	12,491	12,475	12,475	12,475	12,475	12,475	12,475	12,475	12,475	12,475
Margin analysis											
Gross margin	29.0%	29.2%	32.5%	36.1%	36.7%	34.0%	38.4%	40.0%	40.3%	40.9%	39.9%
Operating margin	18.5%	18.5%	21.8%	26.2%	27.1%	23.8%	28.9%	30.8%	31.2%	31.8%	30.7%
Pre-tax margin	20.9%	27.3%	65.7%	20.4%	29.2%	35.1%	31.2%	33.0%	33.2%	33.8%	32.9%
Effective tax rate	16.3%	3.2%	6.6%	14.5%	14.5%	9.1%	15.0%	15.5%	14.5%	14.5%	14.9%
Growth (YoY%)											
Sales	2%	5%	17%	33%	33%	22%	32%	27%	17%	15%	22%
Operating profit	-15%	15%	38%	85%	82%	57%	106%	79%	39%	35%	58%
Net income	-12%	108%	375%	-8%	104%	122%	33%	-43%	90%	33%	7%
EPS	-12%	108%	375%	-8%	104%	122%	33%	-43%	90%	33%	7%
Product mix (%)											
Wafer	98%	95%	97%	97%	97%	97%	97%	97%	97%	96%	97%
Others	2%	5%	3%	3%	3%	3%	3%	3%	3%	4%	3%

Sources: Company data, GF Securities (Hong Kong) Brokerage

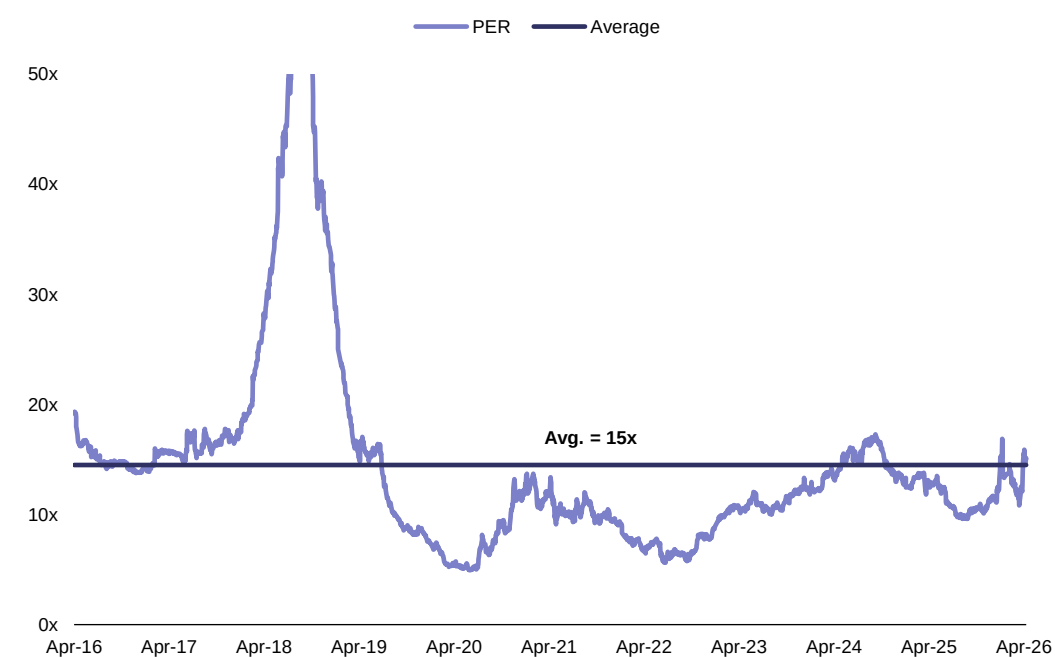
Valuation and Recommendation

Buy with TP of NT\$160

We rate UMC (2303 TT, UMC US) Buy and set our TP at NT\$160, which is based on 20x our 2027E EPS estimate from previous 18x due to enhanced margin outlook. We adopt P/E methodology as we believe the market will be mainly focused on the company's earnings outlook, and its earnings volatility has reduced since 2021 due to a much more diversified application mix and specialty position. We adopt the 2027 P/E valuation is due to the improving pricing environment in the longer term. Overall, we believe the mature node industry outlook has become more favorable, and UMC's mgmt. also guided a favorable pricing environment in 2026/2027.

UMC P/E traded at an average of 15x since 2021, when the market accepted the P/E valuation. Our target price is 20% higher than the mid-cycle level, as we believe the rerating catalysts are emerging amid favorable industry supply/demand. Meanwhile, the company continued to gain AI related business (i.e., networking, silicon photonics, advanced packaging).

Figure 5: UMC's 12-month forward P/E



Sources: Company data, GF Securities (Hong Kong) Brokerage

Risks

1) Geo-political risk; 2) Demand uncertainties; 3) Competition; 4) New product delay.

Balance Sheet						NTD mn
	2024	2025	2026E	2027E	2028E	
Current assets	189,678	204,783	249,471	258,370	330,783	
Cash and cash equivalents	115,239	128,365	159,280	153,153	211,191	
Inventory	35,782	37,228	42,314	46,878	52,476	
Accounts Receivable	33,343	31,274	38,208	46,555	53,561	
Other current assets	5,313	7,915	9,670	11,783	13,556	
Non-current assets	380,523	374,213	377,414	405,364	426,298	
Long-term equity investment	72,487	75,373	78,373	81,373	84,373	
Fixed assets	279,059	271,395	271,597	296,546	314,481	
Other long-term assets	28,976	27,445	27,445	27,445	27,445	
Total assets	570,201	578,996	626,885	663,734	757,081	
Current liabilities	75,260	87,598	91,841	105,567	108,238	
Accounts Payable	7,633	9,170	10,422	11,547	12,926	
Short-term borrowings	19,510	27,597	19,318	18,352	8,258	
Other current liabilities	48,117	50,831	62,100	75,669	87,054	
Non-current liabilities	75,260	87,598	91,841	105,567	108,238	
Long-term Debt	55,533	45,372	31,760	30,172	13,578	
Other non-current liabilities	61,222	66,170	66,170	66,170	66,170	
Total liabilities	192,016	199,141	189,771	201,910	187,986	
Total Equity	378,185	379,855	437,114	461,824	569,095	
Share Capital	125,607	125,882	125,882	125,882	125,882	
Additional Paid-in Capital	25,473	21,004	15,004	6,004	-7,496	
Retained earnings	226,849	232,883	296,141	329,851	450,623	
Preferred Stocks	257	87	87	87	87	
Total Liabilities & Equity	570,201	578,996	626,885	663,734	757,081	

Income Statement						NTD mn
	2024	2025	2026E	2027E	2028E	
Revenue	232,303	237,553	290,217	353,625	406,835	
Cost of sales	156,649	168,647	191,684	212,361	237,721	
Gross profit	75,654	68,906	98,533	141,264	169,114	
Operating Expense	24,042	24,958	29,519	32,558	35,888	
Operating profit	51,613	43,949	69,014	108,706	133,226	
Interest Income	3,670	2,279	1,821	2,359	3,246	
Interest Expense	1,679	1,602	31,297	5,200	5,200	
Net other Non-op. Income/(Loss)	2,616	5,022	-305	0	0	
Pre-tax profit	56,220	49,648	101,828	116,265	141,672	
Income tax	9,218	8,294	9,451	17,415	21,188	
Profit for the year	47,002	41,353	92,376	98,850	120,484	
Minority interest	105	182	162	144	144	
Net profit to ord. equity	47,211	41,716	92,701	99,138	120,772	
EPS (NT\$)	3.8	3.3	7.4	7.9	9.7	

Cash Flow Statement						NTD mn
	2024	2025	2026E	2027E	2028E	
Operating cash flow	79,970	103,286	154,648	168,955	196,326	
Profit for the year	47,211	41,716	92,701	99,138	120,772	
Depreciation & amortization	48,168	59,259	63,199	70,151	77,166	
Change in working capital	-16,251	2,272	-1,252	-333	-1,612	
Others	843	39	0	0	0	
Investing cash flow	-87,144	-52,988	-63,400	-95,100	-95,100	
Capex	-85,408	-48,764	-63,400	-95,100	-95,100	
Change in investment	924	-2,924	0	0	0	
Others	-2,660	-1,300	0	0	0	
Free cash flow	-5,438	54,523	91,248	73,855	101,226	
Financing cash flow	6,135	2,874	-21,891	-2,554	-26,688	
Change in Capital	0	0	0	0	0	
Net Change in Debt	271	-2,075	-21,891	-2,554	-26,688	
Others	5,864	4,948	0	0	0	
Exchange influence	3,715	-1,846	0	0	0	
Net increase in cash	-1,038	53,172	69,357	71,301	74,537	

Key Financial Ratios					
	2024	2025	2026E	2027E	2028E
Growth (%)					
Revenue growth	4.4	2.3	22.2	21.8	15.0
Operating profit growth	-10.8	-14.8	57.0	57.5	22.6
Net profit growth	-7.8	5.2	144.1	25.4	42.9
Profitability (%)					
Gross profit margin	32.6	29.0	34.0	39.9	41.6
Operating Profit Margin	22.2	18.5	23.8	30.7	32.7
Net profit margin	20.3	17.6	31.9	28.0	29.7
Key Ratio (%)					
ROA	8.3	7.2	14.8	14.9	16.0
ROE	12.5	11.0	21.2	21.5	21.2
Stability					
Gross debt/equity	0.5	0.5	0.4	0.4	0.3
Interest Coverage	48	64	115	157	258
Current Ratio	2.5	2.3	2.7	2.4	3.1
Quick Ratio	2.0	1.9	2.3	2.0	2.6
Net debt/equity	Net Cash	Net Cash	Net Cash	Net Cash	Net Cash

Rating definitions Benchmark: Hang Seng Index (Hong Kong)

Company ratings	Buy	Stock expected to outperform benchmark by more than 10%
	Hold	Expected stock relative performance ranges between -10% and 10%
	Underperform	Stock expected to underperform benchmark by more than 10%

Sector ratings	Positive	Sector expected to outperform benchmark by more than 10%
	Neutral	Expected sector relative performance ranges between -10% and 10%
	Cautious	Sector expected to underperform benchmark by more than 10%

Hong Kong	Company	GF Securities (Hong Kong) Brokerage Limited
	Address	27/F, GF Tower, 81 Lockhart Road, Wan Chai, Hong Kong
	Telephone	(852) 37191111
	Email	evanlee@gfgroup.com.hk

Disclaimer

This report has been prepared by GF Securities (Hong Kong) Brokerage Limited ("GF Securities (Hong Kong) Brokerage"). According to the laws, regulations and regulatory requirements in different countries and regions, this report is distributed by GF Securities (Hong Kong) Brokerage with relevant legal and compliant operation qualifications in these countries and regions.

GF Securities (Hong Kong) Brokerage is licensed by the Securities and Futures Commission of Hong Kong ("SFC") to conduct Type 4 Regulated Activity "Advising on Securities". It is regulated by the SFC, and is responsible for the distribution of this report in Hong Kong. Information about the qualifications of the research analyst(s) who is(are) the author(s) of this report as licensed by the SFC are disclosed in the section where research analyst names are shown.

The research analyst(s) primarily responsible for the content of this report, in whole or in part, certifies that with respect to the company or relevant securities that the analyst(s) covered in this report: 1) all of the views expressed accurately reflect his or her personal views on the company or relevant securities mentioned herein; and 2) no part of his or her remuneration was, is, or will be, directly or indirectly, in connection with his or her specific recommendations or views expressed in this report.

This report is published solely for information purpose and does not constitute an offer to buy or sell any securities or a solicitation of an offer to buy, or recommendation for investment in, any securities.

The securities mentioned in this report may not be allowed to be sold in certain jurisdictions. No action has been taken to permit the distribution of the research report to any person in any jurisdiction that the circulation or distribution of such research report is unlawful. This report is distributed solely to clients or designated institutions authorized by GF Securities (Hong Kong) Brokerage, and is not distributed publicly. It is distributed to certain clients based on the conclusion that they are able to assess investment risks independently, execute investment decisions independently and assume corresponding risks independently.

This report has been issued and based on information obtained from sources generally available to the public and believed by the research analyst(s) to be reliable but which has not been independently verified. No representation or warranty, either express or implied, is made by GF Securities (Hong Kong) Brokerage as to their accuracy and completeness of the information contained in this report. GF Securities (Hong Kong) Brokerage accepts no liability for all loss arising from the use of the materials presented in this report, unless is excluded by applicable laws or regulations. Please be aware of the fact that investments involve risks and the price of securities may be fluctuated and therefore return may be varied, past results do not guarantee future performance. Any recommendation contained in this report does not have regard to the specific investment objectives, financial situation and the particular needs of any individuals. This report is not to be taken in substitution for the exercise of judgment by respective recipients of this report, where necessary, recipients should obtain professional advice before making investment decisions.

GF Securities (Hong Kong) Brokerage may have issued, and may in the future issue, other communications that are inconsistent with, and reach different conclusions from, the information presented in the research report. The points of view, opinions and analytical methods adopted in the research report are solely expressed by the analysts but not that of GF Securities (Hong Kong) Brokerage or its affiliates. The information, opinions and forecasts presented in the research report are the current opinions of the analysts as of the date appearing on this material only which may subject to change at any time without notice. The salesperson, dealer or other professionals of GF Securities (Hong Kong) Brokerage may deliver opposite points of view to their clients and the proprietary trading division with respect to market commentary or dealing strategy either in writing or verbally. The proprietary trading division of GF Securities (Hong Kong) Brokerage may have different investment decision which may be contrary to the opinions expressed in the research report. GF Securities (Hong Kong) Brokerage or its affiliates or respective directors, officers, analysts and employees related to research report business may have rights and interests in securities mentioned in the research report. Recipients should be aware of relevant disclosure of interest (if any) when reading this report.

GF Securities (Hong Kong) Brokerage and its affiliates may be seeking or building business relationships with company(ies) mentioned in this report. Therefore, investors should consider the impact on the independence of this report by GF Securities (Hong Kong) Brokerage and its affiliates due to potential conflicts of interests. Investors should not make any investment decisions based solely on the contents of this report. Investors should make their own investment decisions and bear their own risk. No written or verbal commitment of sharing gains or losses from securities investments in any form shall be effective.

This report may contain and/or describe/present factual historical information on prices of Futures contracts (the "information"). Please note that this information is solely for the purpose of forming part of the argument/grounds/evidence in our research methodology/analysis to support our conclusion on our view of the relevant industry/company mentioned. It does not, by any means (express or implied) to be associated with or constituted as SFC Type 5 Regulated Activity (Advising on futures contracts).

Disclosure of Interests

- 1) The research analyst(s) and his/her associate has not served as an officer of the company(ies) mentioned in this report, and does not have any financial interests in the company(ies) mentioned in this report.
- 2) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any financial interests and has not invested in interests aggregate to an amount equal to or more than (i) 1% of the market capitalization; or (ii) 1% of the issued share capital, or issued units, in the company(ies) mentioned in this report.
- 3) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any market making activities, and has not employed any individual(s) serving as officer(s) of the company(ies) mentioned in this report.
- 4) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any investment banking relationship in Hong Kong with the company(ies) mentioned in this report in the past 12 months.

Copyright © GF Securities (Hong Kong) Brokerage

Without the prior written consent obtained from GF Securities (Hong Kong) Brokerage, any part of the materials contained herein should not (i) in any forms be copied or reproduced or (ii) be re-disseminated.